24 There are

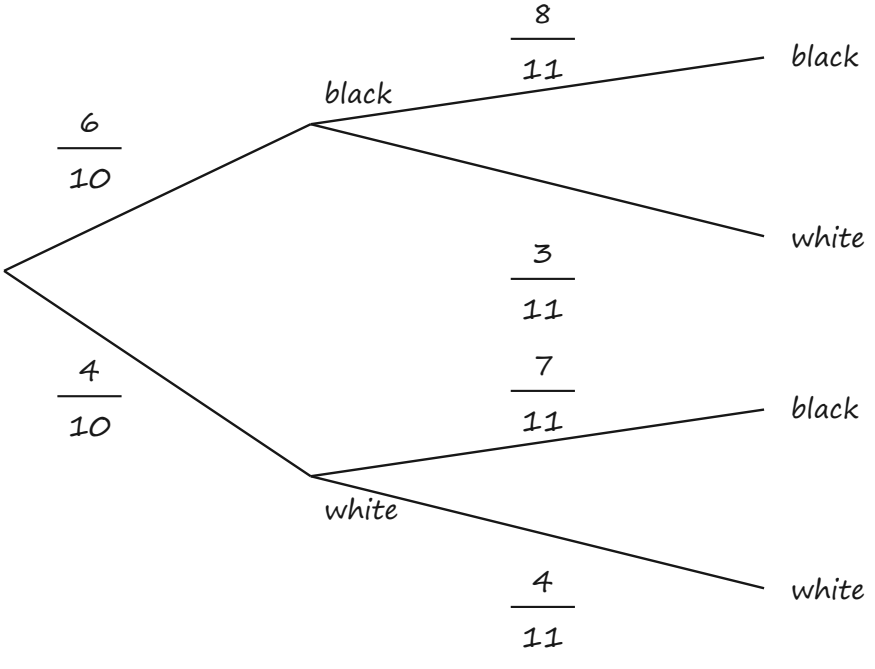
6 black counters and 4 white counters in bag **A**
7 black counters and 3 white counters in bag **B**
5 black counters and 5 white counters in bag **C**

Bernie takes at random a counter from bag **A** and puts the counter in bag **B**.
He then takes at random a counter from bag **B** and puts the counter in bag **C**.

Find the probability that there are now more black counters than white counters in bag **C**.

Bag C has more black counters exactly when a black counter is transferred from bag B.

A → B B → C



1 mark
Multiply along a successful path

$$P = \frac{6}{10} \times \frac{8}{11} + \frac{4}{10} \times \frac{7}{11}$$

1 mark
Add both successful paths

$$= \frac{48 + 28}{110} = \frac{76}{110} = \frac{38}{55}$$

1 mark
38/55

(Total for Question 24 is 3 marks)